Nikos Tsikouras

Machine Learning Researcher

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PROFILE

PhD researcher with first-author papers at **NeurIPS** and **ICLR**, including a **NeurIPS** Oral. I develop algorithms with provable guarantees for representation learning, dimensionality reduction, and randomized methods; current work targets LLM compression and efficient inference. Seeking ML research and applied scientist internships for Summer 2027.

PUBLICATIONS

Accepted

- MaxSketch: Robust Distinct Counting in Streams via Random Projections — **NeurIPS 2026 (Oral)**
  N. Tsikouras, C. Caramanis, C. Tzamos
- A Derandomization Framework for Structure Discovery: Applications in Neural Networks and Beyond — **ICLR 2026**
  N. Tsikouras, Y. Pantis, I. Mitliagkas, C. Tzamos
- Optimization Can Learn Johnson–Lindenstrauss Embeddings — **NeurIPS 2024**
  N. Tsikouras, C. Caramanis, C. Tzamos

Under Review (first author; co-authors omitted for anonymity)

- SGD Selects Initialization-Dependent Representation in Overparameterized Networks — **ICLR 2027**
- Beyond Labels: PAC Learning with Intermediate Supervision — **ICLR 2027**

RESEARCH EXPERIENCE & PROJECTS

PhD Researcher – Machine Learning 2023–Present

University of Athens & Archimedes, Athena Research Center — Athens, Greece

- Currently investigating theory-inspired approaches to training-aware quantization and compression of large language models.
- Advisors: Christos Tzamos (NKUA) and Constantine Caramanis (UT Austin).
- Research visits: **MIT** with Constantinos Daskalakis (2026); **UT Austin** with Constantine Caramanis (2025).

ReasonScale – LLM Reasoning Evaluation 2026

Personal Project

- Built a Python framework to evaluate LLMs on mathematical reasoning tasks, integrating GSM8K, Hugging Face Datasets, and the Groq API.
- Developed tools to extract and verify numerical answers from model outputs, with automated testing to ensure reliable evaluation.
- Designed and evaluated adaptive test-time compute strategies for LLM reasoning, using prediction uncertainty to allocate inference budgets and optimize accuracy–cost trade-offs.

EDUCATION

PhD in Machine Learning, University of Athens — 2023–Present
Expected graduation: 2028

MRes in Statistics, Lancaster University — 2021–2022
Distinction

MSc in Statistics, Imperial College London — 2020–2021
Distinction; Data Science and Machine Learning specialization

BSc in Mathematics, University of Patras — 2016–2020
Valedictorian; 9.19/10

TECHNICAL & ADDITIONAL

Programming	Python, C++, R; Git
ML / LLM Tools	PyTorch, Hugging Face, scikit-learn; LLM APIs, evaluation pipelines
Co-Founder	Plateia AI – Nonprofit initiative promoting AI literacy, public engagement, and responsible AI in Greece
Leadership	Team leader and coach, Greek delegations to the International and European AI Olympiads (2025–2026); students earned 4 silver and 4 bronze medals
Service	Reviewer: NeurIPS, ICML, ICLR; invited speaker, Archimedes Conference on AI and Algorithms (2024, 2025)
Selected Awards	4-year doctoral funding; Latsis Foundation scholarship for MSc studies at Imperial College London